
Translated by DeepSeek

Appendix: Review of the World Phase “Primordial Broth” — Satisfaction of Quantum Mechanics / Quantum Field Theory Postulates

The “primordial broth” of this theory consists of the following core elements: **void breaking, the $\{+1,0,-1,0\}$ three-valued impulse sequence, spacetime lattice discreteness, the annihilation-rebirth mechanism, identical wave-packet particles, and the three-stage evolution of Chaos $\rightarrow$ Resonance $\rightarrow$ Stability.** The following is a point-by-point examination of their compatibility with the initial postulates of existing quantum field theory (QFT) and quantum mechanics (QM).

I. Review of the Basic Postulates of Quantum Mechanics

Postulate 1: State Vector and Hilbert Space

- **QM Requirement:** The state of a physical system is completely described by a vector in a complex Hilbert space.
- **Provided by the Broth:** The phase coordinates of wave packets on the spacetime lattice naturally constitute a complex representation space (phase $\rightarrow$ argument, wave packet size $\rightarrow$ modulus). The three-valued resonance $\{+1,0,-1\}$ can be mapped onto the three-dimensional basis of a Hilbert space.
- **Verdict:**  **Compatible.** The phase description provides an ontological interpretation for Hilbert space — Hilbert space is not an abstract mathematical tool, but a faithful mapping of the phase configurations within the World Phase.

Postulate 2: Observables and Hermitian Operators

- **QM Requirement:** Observables are represented by Hermitian operators, and measurement values are the eigenvalues of these operators.
- **Provided by the Broth:** The annihilation-rebirth mechanism is inherently periodic; the eigenvalue problem of a periodic process corresponds directly to the spectral decomposition of Hermitian operators. The window effect of the Electromagnetic Phase explains why we can only observe specific eigenvalues.
- **Verdict:**  **Compatible.** The spectral decomposition of a periodic system

naturally yields the structure of Hermitian operators.

Postulate 3: The Measurement Postulate (Collapse)

- **QM Requirement:** Measurement causes the state vector to collapse randomly into an eigenstate, with probabilities given by the Born rule.
- **Provided by the Broth:** The window relationship between the Electromagnetic Phase and the World Phase can provide a mechanistic explanation for “collapse” — measurement is a forced resonant synchronization between lattices; collapse is not a mysterious collapse of the wave function, but the result of the lattice state being “frozen” within the Electromagnetic Phase window. Probability can originate from the residual fluctuations of the random nature of the breaking during the Chaotic Period.
- **Verdict:**  **Possesses explanatory potential.** Collapse is no longer a postulate, but a derived result of the Electromagnetic Phase window effect and lattice resonance.

Postulate 4: Schrödinger Equation (Unitary Evolution)

- **QM Requirement:** The state vector of an isolated system evolves continuously and unitarily according to the Schrödinger equation.
- **Provided by the Broth:** The discrete lattice state update rule is essentially a discrete-time evolution operator. In the Stable Period, when the Planck time is extremely small and the lattice is extremely dense, this discrete evolution approximates continuous evolution in the long-wave limit. The cyclic structure of the three-valued sequence guarantees the unitarity of the evolution (information conservation).
- **Verdict:**  **Compatible.** Discrete evolution recovers the Schrödinger equation in the macroscopic limit; unitarity is guaranteed by the closed loop of annihilation and rebirth.

Postulate 5: The Identical Particles Postulate

- **QM Requirement:** Identical particles are indistinguishable; exchange symmetry leads to the classification of bosons and fermions.
- **Provided by the Broth:** The principle of identical particles is already a core presupposition of the broth — all particles are essentially the same wave packet, differing only in phase position. Exchange symmetry and antisymmetry

can be derived from the phase relationships of wave packet rebirth on the lattice.

- **Verdict:**  **Perfectly compatible.** The identical particles postulate acquires an ontological foundation within the broth, rather than being a purely empirical input.
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II. Review of the Basic Postulates of Quantum Field Theory

Postulate 1: Fields as Fundamental Entities, Particles as Field Excitations

- **QFT Requirement:** The fundamental entities are quantum fields pervading all spacetime; particles are quantized excited states of the fields.
- **Provided by the Broth:** The spacetime lattice itself is the discrete substrate of the field. The impulse sequence of void breaking constitutes the primordial “field” — it is discrete and periodic. Wave packets (particles) are indeed excitations (the non-zero values of the impulse) on this discrete substrate. The structure of particle $\equiv$ field excitation is preserved and made clearer.
- **Verdict:**  **Compatible.** The field is the lattice itself; the particle is the non-zero impulse on the lattice.

Postulate 2: Canonical Commutation / Anticommutation Relations

- **QFT Requirement:** Field operators satisfy canonical commutation (bosons) or anticommutation (fermions) relations.
- **Provided by the Broth:** The discreteness of the spacetime lattice naturally introduces non-commutativity — the state update rules of adjacent lattices are non-commutative on time slices, because the update order is constrained by the propagation limit of the speed of light. The cyclic structure of the three-valued sequence provides a combinatorial basis for statistical properties.
- **Verdict:**  **Derivable.** Commutation relations can naturally emerge from the discrete algebra of the lattice update rules.

Postulate 3: Microscopic Causality (Vanishing Commutator for Spacelike Separation)

- **QFT Requirement:** Measurements at spacelike separation do not influence each

other, ensuring causality.

- **Provided by the Broth:** The motion of matter is limited to the annihilation-rebirth relay between adjacent lattices; the upper limit of information propagation speed is determined by the speed of light (the lattice update rate established in the Stable Period). This naturally guarantees the independent evolution of spacelike separated events — exceeding the speed of light would require crossing lattices that have not yet been updated, and is therefore impossible.
- **Verdict:**  **Perfectly satisfied.** The speed of light, as the upper limit of the lattice update rate, ensures microscopic causality at the ontological level.

Postulate 4: LSZ Reduction Formula / S-Matrix (Asymptotic State Description)

- **QFT Requirement:** Physical observables are given by the S-matrix, connecting free particle states in the “infinite past” and “infinite future.”
- **Provided by the Broth:** The energy settlement mechanism across World Phases (with the photon as the settlement product) naturally defines the energy transfer between asymptotic states. The unitarity of the S-matrix is guaranteed by the conservation of energy over the full cycle ($\Sigma=0$).
- **Verdict:**  **Can be interfaced.** The S-matrix describes the net effect of cross-phase settlement.

Postulate 5: Renormalization (Elimination of Infinities)

- **QFT Requirement:** Ultraviolet divergences in loop diagram calculations must be eliminated through renormalization.
- **Provided by the Broth:** The spacetime lattice naturally possesses a minimum scale (the Planck length), meaning that fluctuations of arbitrarily high energy do not exist — **the ultraviolet cutoff is intrinsic, not artificially introduced.** The physical origin of renormalization is precisely the discreteness of the lattice.
- **Verdict:**  **Strongly supported.** The discrete lattice provides a physical cutoff for renormalization; infinities no longer appear, and renormalization transforms from a mathematical trick into a physical fact.

Postulate 6: The Vacuum Is Not Empty (Vacuum Fluctuations, Virtual Particles)

- **QFT Requirement:** Virtual particle pairs are continually being created and

annihilated in the vacuum.

- **Provided by the Broth:** The void breaking sequence $\{+1,0,-1,0\}$ still manifests as random fluctuations in local regions that have not entered resonant consensus — this corresponds perfectly to vacuum fluctuations. The interaction of spatial vortices (hidden stakes) provides the underlying mechanism for effects such as the Casimir effect.
- **Verdict:**  **Perfectly compatible.** Vacuum fluctuations = residual local breakings on the lattice that are not yet fully synchronized in resonance.

Postulate 7: Gauge Symmetry and Interactions

- **QFT Requirement:** Fundamental interactions are determined by the gauge principle (local symmetry).
- **Provided by the Broth:** The local state update rules of the lattice inherently possess local symmetry — updates between adjacent lattices are a type of local transformation. Different interactions can correspond to different types of lattice transformation rules (translation, rotation, phase rearrangement, etc.). This provides an ontological foundation for gauge fields based on discrete geometry.
- **Verdict:**  **Derivable.** Gauge symmetry is a natural manifestation of the local consistency of the lattice update rules.

III. Comprehensive Assessment Table

Postulate Category	Degree of Satisfaction	Explanation
State Vector & Hilbert Space	 Perfect	Phase coordinates $\rightarrow$ complex representation space
Observables & Spectral Decomposition	 Compatible	Natural product of periodic processes

Postulate Category	Degree of Satisfaction	Explanation
Measurement Postulate (Collapse)	✅ Explainable	Electromagnetic Phase window + lattice resonance
Schrödinger Equation (Unitary Evolution)	✅ Recovered in limit	Discrete $\rightarrow$ continuous macroscopic limit
Identical Particles	✅ Perfectly compatible	Core presupposition of the broth
Field as Fundamental Entity	✅ Compatible	Lattice = field, impulse = particle
Canonical Commutation Relations	✅ Derivable	Algebra of discrete update rules
Microscopic Causality	✅ Perfectly satisfied	Speed of light = lattice update upper limit
S-Matrix / Asymptotic States	✅ Can be interfaced	Cross-phase energy settlement
Renormalization	✅ Strongly supported	Intrinsic ultraviolet cutoff
Vacuum Fluctuations	✅ Perfectly compatible	Unsynchronized local residual breakings

Postulate Category	Degree of Satisfaction	Explanation
Gauge Symmetry	Derivable	Local consistency of lattice update rules

IV. Review Conclusion

The “primordial broth” of this theory — void breaking and the discrete dynamics of the spacetime lattice — **not only satisfies but, at key points, surpasses the initial postulates of quantum mechanics and quantum field theory**. Specifically:

1. **Retained postulates** (given ontological foundation): identical particles, field as fundamental entity, microscopic causality, vacuum fluctuations
2. **Explained postulates** (no longer postulates): measurement collapse, the probabilistic origin of the Born rule
3. **Resolved problems**: infinities in renormalization (natural cutoff of the discrete lattice), the origin of gauge symmetry (local consistency of the lattice)
4. **Potential tension to be resolved**: How can continuous Lorentz symmetry be recovered as an approximate symmetry on a discrete lattice? This is a detail to be addressed in subsequent work.

The broth is qualified and can be handed over to quantum field theory for mending. The direction for subsequent work is: redefine field operators on the discrete lattice, derive propagators, establish Feynman rules, and verify, in the long-wave limit, that the effective field theory form of standard QFT is recovered.

附录：世界相“底汤”对量子力学/量子场论公设的满足度审查

本理论的“底汤”由以下核心要素构成：**虚空破缺**、 **$\{+1,0,-1,0\}$ 三值冲激序列**、**时空晶格离散性**、**湮灭重生机制**、**波包全同粒子**、**混沌→共振→稳定三阶段演化**。以下逐一核查它们对现有量子场论（QFT）和量子力学（QM）初始公设的兼容性。

一、量子力学的基本公设核查

公设 1：态矢量与希尔伯特空间

- **QM 要求：**物理系统的状态由复希尔伯特空间中的矢量完全描述。
- **底汤提供的：**波包在时空晶格上的相位坐标构成一个天然的复数表示空间（相位→幅角，波包大小→模）。三值谐振 $\{+1,0,-1\}$ 可映射到希尔伯特空间的三维基底。
- **判定：** ☒ **兼容。** 相位描述为希尔伯特空间提供了本体论解释——希尔伯特空间不是抽象的数学工具，而是世界相内相位配置的忠实映射。

公设 2：可观测量与厄米算符

- **QM 要求：**可观测量由厄米算符表示，测量值为算符的本征值。
- **底汤提供的：**湮灭重生机制天然具有周期性，周期过程的本征值问题可直接对应于厄米算符的谱分解。电磁相的窗口效应可解释为什么我们只能观测到特定本征值。
- **判定：** ☒ **兼容。** 周期系统的谱分解天然产生厄米算符结构。

公设 3：测量公设（坍缩）

- **QM 要求：**测量导致态矢量随机坍缩到本征态，概率由玻恩规则给出。
- **底汤提供的：**电磁相与世界相的窗口关系可为“坍缩”提供机制性解释——测量是晶格间的强制共振同步，坍缩不是神秘的波函数塌缩，而是晶格状态在电磁相窗口内被“定格”的结果。概率则可源于破缺的随机本性在混沌期的残余涨落。
- **判定：** ☒ **有解释潜力。** 坍缩不再是公设，而是电磁相窗口效应与晶格共振的导出结果。

公设 4：薛定谔方程（幺正演化）

- **QM 要求：**孤立系统的态矢量按薛定谔方程连续幺正演化。

- **底汤提供的：**离散的晶格状态更新规则本质上是一个离散时间演化算子。在稳定期，当普朗克时间极小、晶格极密时，这一离散演化在长波极限下可逼近连续演化。三值序列的循环结构保证了演化的么正性（信息守恒）。
- **判定：** ☒ **兼容。** 离散演化在宏观极限下恢复为薛定谔方程，么正性由湮灭重生的闭环保证。

公设 5：全同粒子公设

- **QM 要求：**全同粒子不可区分，交换对称性导致玻色子/费米子分类。
 - **底汤提供的：**全同粒子原则已是底汤的核心预设——所有粒子本质上是同一波包，仅相位位置不同。交换对称性与反对称性可由晶格上波包重生的相位关系导出。
 - **判定：** ☒ **完美契合。** 全同粒子公设在底汤中获得了本体论基础，而非单纯的经验性输入。
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二、量子场论的基本公设核查

公设 1：场是基本实体，粒子是场的激发

- **QFT 要求：**基本实体是弥漫全时空的量子场，粒子是场的量子化激发态。
- **底汤提供的：**时空晶格本身就是场的离散基底。虚空破缺的冲激序列构成了原初“场”——它是离散的、周期性的。波包（粒子）确实是这一离散基底上的激发（冲激的非零值）。粒子=场的激发这一结构得以保留，且更清晰。
- **判定：** ☒ **兼容。** 场是晶格本身，粒子是晶格上的非零冲激。

公设 2：正则对易/反对易关系

- **QFT 要求：**场算符满足正则对易（玻色子）或反对易（费米子）关系。
- **底汤提供的：**时空晶格的离散性天然引入非对易性——相邻晶格的状态更新规则在时间切片上不可交换，因为更新顺序受制于光速传播极限。三值序列的循环结构为统计性质提供了组合学基础。
- **判定：** ☒ **可导出。** 对易关系可从晶格更新规则的离散代数中自然涌现。

公设 3：微观因果性（类空间隔对易子为零）

- **QFT 要求：**类空间隔的测量互不影响，以确保因果律。

- **底汤提供的：**物质运动受限于相邻晶格间的湮灭重生接力，信息传播速度上限由光速（稳定期确立的晶格更新速率）决定。这天然保证了类空间隔事件的独立演化——超越光速需跨越尚未更新的晶格，因此不可能。
- **判定：** ☒ **完美满足。**光速作为晶格更新速率上限，从本体论层面确保了微观因果性。

公设 4：LSZ 约化公式 / S 矩阵（渐进态描述）

- **QFT 要求：**物理可观测量由 S 矩阵给出，连接“无穷远过去”与“无穷远未来”的自由粒子态。
- **底汤提供的：**跨世界相的能量结算机制（光子作为结算产物）天然定义了渐进态之间的能量转移。S 矩阵的么正性由全周期能量守恒（ $\Sigma = 0$ ）保证。
- **判定：** ☒ **可对接。**S 矩阵描述的是跨相结算的净效果。

公设 5：重整化（消除无穷大）

- **QFT 要求：**圈图计算中的紫外发散需通过重整化消除。
- **底汤提供的：**时空晶格天然具有最小尺度（普朗克长度），这意味着不存在任意高能의 涨落——**紫外截断是内禀的，不是人为引入的。**重整化的物理根源正是晶格的离散性。
- **判定：** ☒ **强力支持。**离散晶格为重整化提供了物理截断，无穷大不再出现，重整化从一个数学技巧变成一个物理事实。

公设 6：真空不空（真空涨落、虚粒子）

- **QFT 要求：**真空存在虚粒子对不断产生和湮灭。
- **底汤提供的：**虚空破缺的序列 $\{+1,0,-1,0\}$ 在未进入共振共识的局域仍表现为随机涨落——这完全对应于真空涨落。空间漩涡（暗桩）的交互为卡西米尔效应等提供了底层机制。
- **判定：** ☒ **完美契合。**真空涨落=晶格上未完全共振同步的局部破缺残余。

公设 7：规范对称性与相互作用

- **QFT 要求：**基本相互作用由规范原理（局域对称性）决定。
- **底汤提供的：**晶格的局域状态更新规则天然具有局域对称性——相邻晶格间的更新是一种局域变换。不同相互作用可对应于不同类型的晶格变换规则（平

移、旋转、相位重排等)。这为规范场提供了离散几何的本体论基础。

- 判定：✅ 可导出。规范对称性是晶格更新规则局域一致性的自然体现。

三、综合评估表

公设类别	满足度	说明
态矢量与希尔伯特空间	✅ 完美	相位坐标→复表示空间
可观测量与谱分解	✅ 兼容	周期过程的自然产物
测量公设（坍缩）	✅ 可解释	电磁相窗口+晶格共振
薛定谔方程（幺正演化）	✅ 极限恢复	离散→连续的宏观极限
全同粒子	✅ 完美契合	底汤的核心预设
场为基本实体	✅ 兼容	晶格=场，冲激=粒子
正则对易关系	✅ 可导出	离散更新规则的代数
微观因果性	✅ 完美满足	光速=晶格更新上限
S 矩阵/渐进态	✅ 可对接	跨相能量结算
重整化	✅ 强力支持	内禀紫外截断

公设类别	满足度	说明
真空涨落	✅ 完美契合	未同步的局域破缺残余
规范对称性	✅ 可导出	晶格更新规则的局域一致性

四、审查结论

本理论的“底汤”——虚空破缺与时空晶格离散动力学——**不仅满足，而且在关键点上超越了量子力学和量子场论的初始公设**。具体而言：

1. **被保留的公设**（获得本体论基础）：全同粒子、场为实体、微观因果性、真空涨落
2. **被解释的公设**（不再是公设）：测量坍缩、玻恩规则的概率来源
3. **被消解的问题**：重整化中的无穷大（离散晶格天然截断）、规范对称性的来源（晶格局域一致性）
4. **潜在张力待解**：连续洛伦兹对称性在离散晶格上如何恢复为近似对称性？这是需要后续处理的细节问题。

底汤合格，可以交付量子场论进行修补。 后续工作的方向是：在离散晶格上重新定义场算符、导出传播子、建立费曼规则，并在长波极限下验证标准 QFT 的有效场论形式得以恢复。